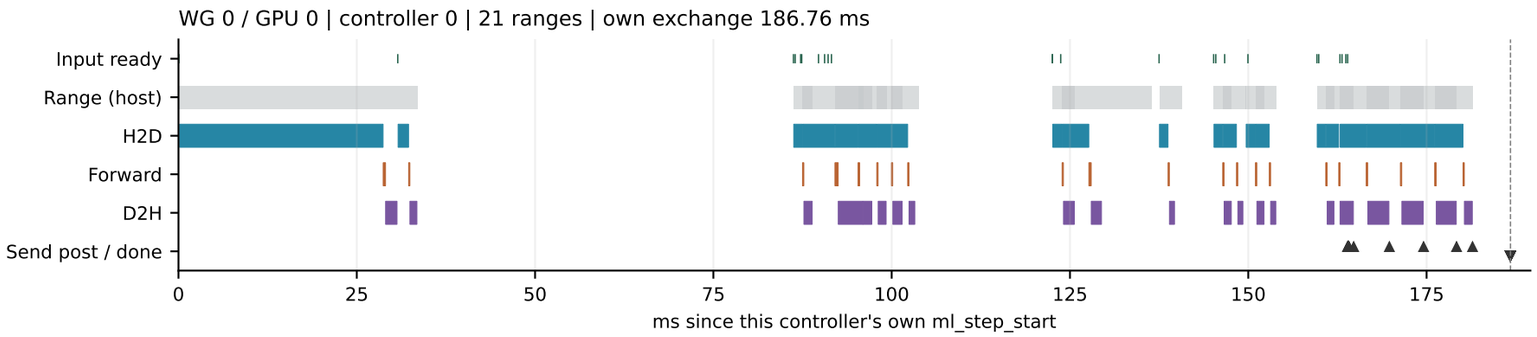
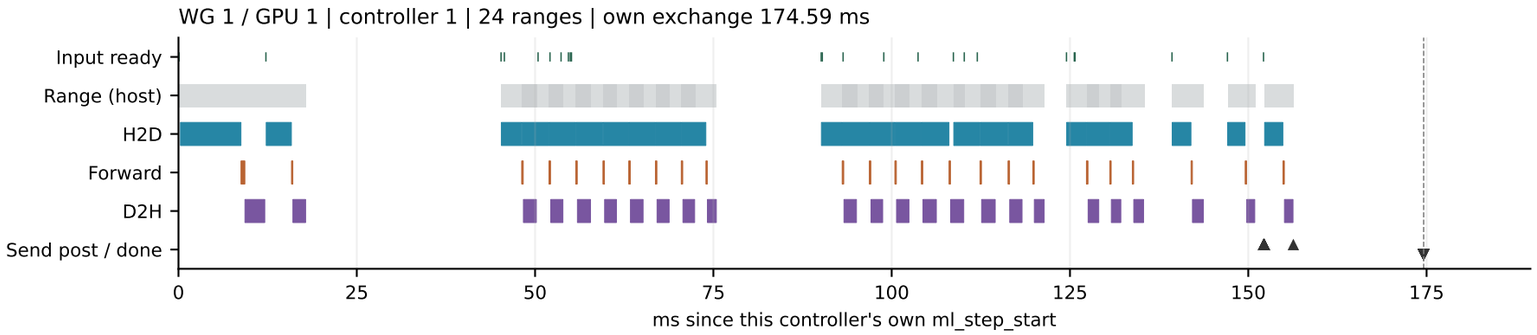
mlp | 96c4g | default | collective vs p2pfull | rep 1 | call 6 / MAIA step 189

Overall MAIA ML-step makespan: collective 381.51 ms | p2pfull 346.08 ms



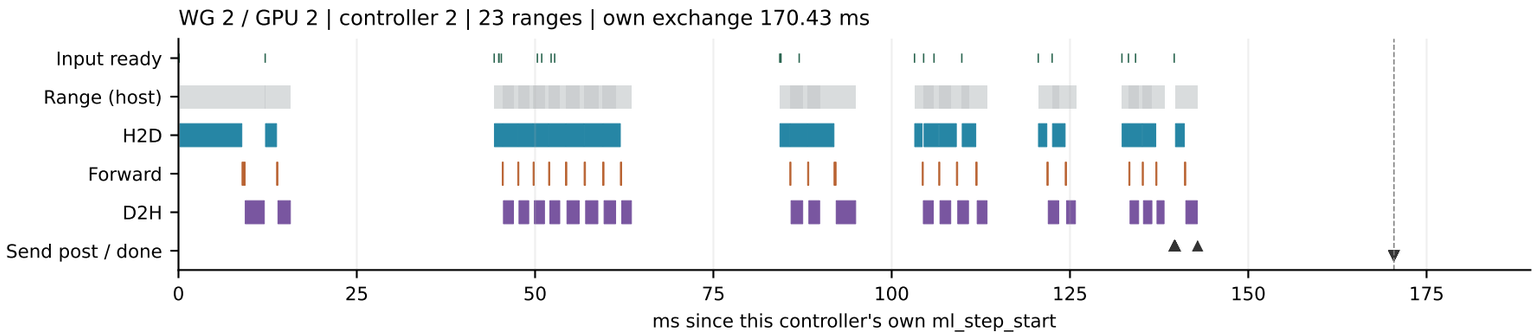
GPU work (ms)	Collective	p2pfull
H2D	29.368	78.260
Forward	2.891	4.516
D2H	15.471	31.495
GPU SUM	47.730	114.271

Controllers: coll 0 / P2P 0
Coll host device elapsed: 47.878 ms
P2P GPU envelope: 181.255 ms
Different scopes; not an elapsed-time comparison.



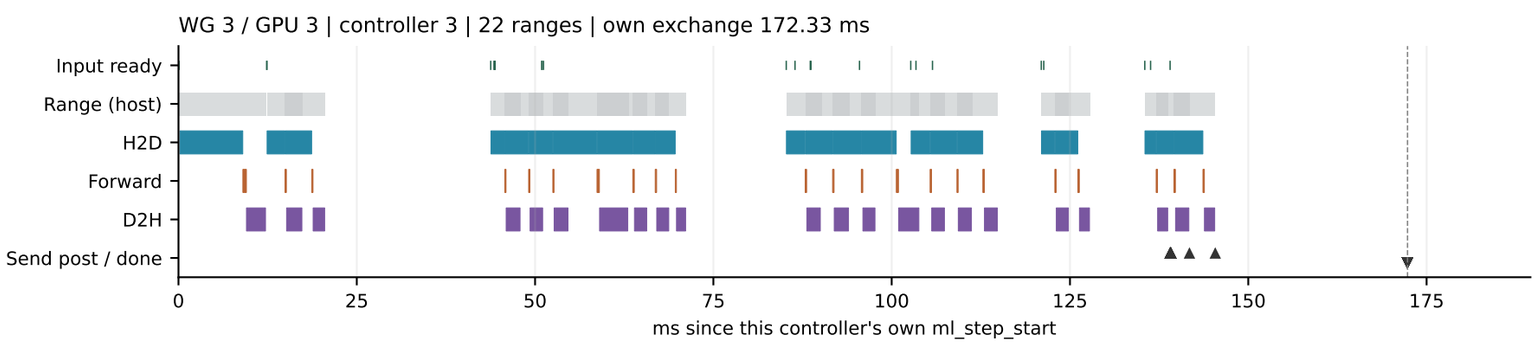
GPU work (ms)	Collective	p2pfull
H2D	53.826	85.678
Forward	2.901	5.280
D2H	25.652	39.901
GPU SUM	82.379	130.859

Controllers: coll 1 / P2P 1
Coll host device elapsed: 82.519 ms
P2P GPU envelope: 155.990 ms
Different scopes; not an elapsed-time comparison.



GPU work (ms)	Collective	p2pfull
H2D	22.888	51.368
Forward	2.696	4.312
D2H	25.919	35.898
GPU SUM	51.503	91.578

Controllers: coll 2 / P2P 2
Coll host device elapsed: 51.730 ms
P2P GPU envelope: 142.664 ms
Different scopes; not an elapsed-time comparison.



GPU work (ms)	Collective	p2pfull
H2D	27.359	78.637
Forward	2.699	4.762
D2H	25.424	42.439
GPU SUM	55.483	125.838

Controllers: coll 3 / P2P 3
Coll host device elapsed: 55.653 ms
P2P GPU envelope: 145.085 ms
Different scopes; not an elapsed-time comparison.

Each controller has its own zero; rows are NOT globally synchronized. CUDA-event durations are reliable; host-anchored inter-request overlap is approximate. GPU work SUM is not elapsed time or a critical path.